

Remote Desktop Services (RDS) RemoteApp Deployment on Microsoft Azure

Project Report

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1. Executive Summary

This project demonstrates the deployment of a **Microsoft Remote Desktop Services (RDS)** environment on **Microsoft Azure**. The solution was designed to publish **WordPad** as a **RemoteApp**, allowing users to access only the application through **RD Web Access**, rather than a complete Remote Desktop session.

The deployment was implemented using a **single Windows Server 2022 virtual machine**, which hosted the required RDS roles to minimize infrastructure cost while satisfying all functional requirements.

2. Project Objective

The objectives of this project were:

- Deploy Remote Desktop Services on Azure.
 - Publish WordPad as a RemoteApp.
 - Provide secure access using RD Web Access.
 - Configure a Public DNS Name.
 - Restrict unnecessary network exposure.
 - Demonstrate successful RemoteApp connectivity.
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3. Solution Architecture

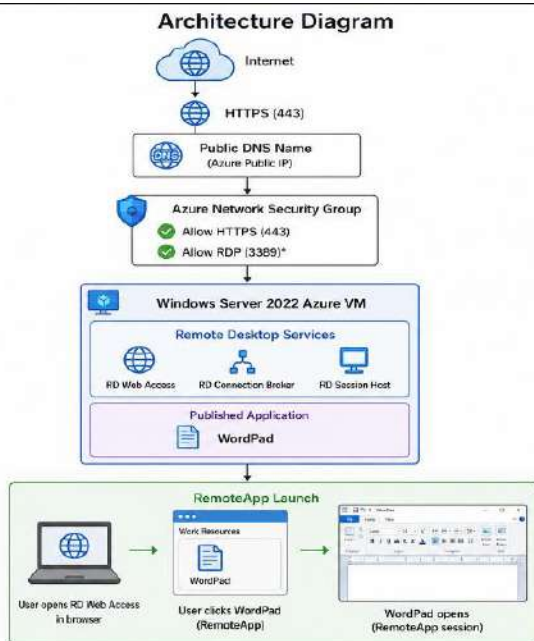
Deployment Model

Single Server Deployment

The following RDS roles were installed on one Windows Server 2022 Virtual Machine:

Role	Purpose
RD Connection Broker	Manages RemoteApp connections
RD Session Host	Hosts published applications
RD Web Access	Provides browser-based application portal

Architecture Diagram



4. Infrastructure Design

Azure Resources

Resource	Configuration
Cloud Platform	Microsoft Azure
Operating System	Windows Server 2022 Datacenter
VM Size	Standard B2s
Region	Central India

Resource	Configuration
Public IP	Static
DNS Label	Configured
Virtual Network	Azure VNet
Network Security Group Enabled	

5. Deployment Steps

Step 1

Created a Windows Server 2022 Virtual Machine.

Step 2

Configured:

- Resource Group
 - Virtual Network
 - Subnet
 - Static Public IP
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Step 3

Installed Remote Desktop Services roles using Server Manager.

Installed Roles:

- RD Session Host
 - RD Connection Broker
 - RD Web Access
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Step 4

Created the default RDS deployment.

Step 5

Created a RemoteApp Collection.

Collection Name: QuickSessionCollection

Step 6

Published RemoteApp.

Application Published: WordPad (write.exe)

6. RemoteApp Configuration

The RemoteApp collection was configured to expose only WordPad.

Published Application:

- WordPad

This ensures users launch only the application instead of receiving a full Windows desktop.

7. Public Access Configuration

Configured Azure Public IP.

Assigned DNS Name:

<https://<DNS-NAME>.centralindia.cloudapp.azure.com/RDWeb>

RD Web Access was successfully accessible through the public URL.

8. Security Configuration

Network Security Group Rules

Port	Protocol	Purpose
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443	TCP	RD Web Access
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3389	TCP	Administrative RDP
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Security Recommendation

RDP (3389) should **not** remain open to the entire Internet.

Best practice:

- Restrict RDP to administrator IP
 - Use Azure Bastion where possible
 - Enable Azure NSG protection
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9. Testing and Validation

Test 1

Verified RD Web Access.

Result:**PASS**

Test 2

Authenticated using Administrator credentials.

Result:**PASS**

Test 3

Verified published applications.

Applications displayed:

- WordPad
- Paint
- Calculator

Test 4

Attempted RemoteApp launch.

Expected Result:

Only WordPad should open.

Observed Result:

RemoteApp launch encountered a Connection Broker configuration issue.

The downloaded RDP file referenced the internal server address:

RDS-SERVER.LAB.LOCAL

instead of the Azure Public DNS name.

This prevented successful RemoteApp launch from an external client.

10. Challenges Encountered

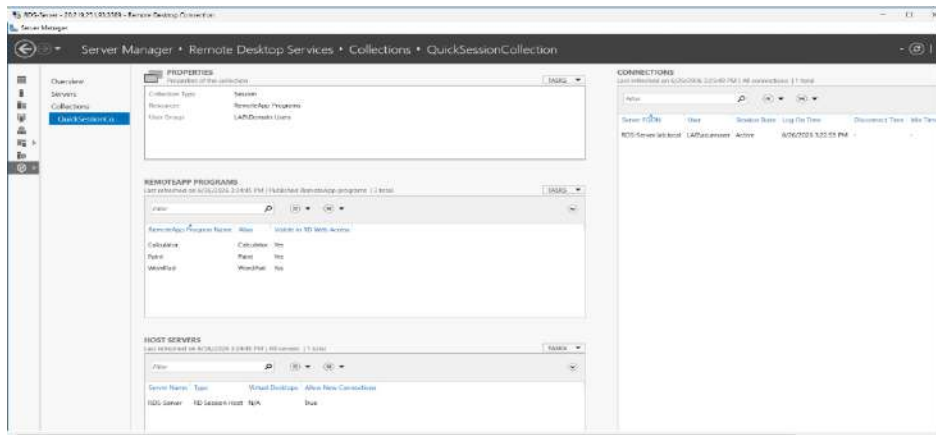
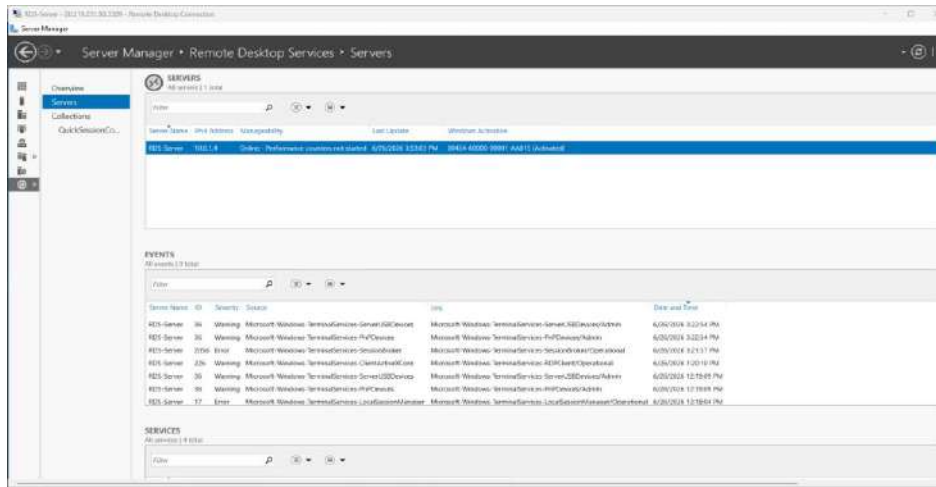
During testing, the following issues were identified:

- RD Web portal was accessible.
- Authentication completed successfully.
- RemoteApp icons appeared correctly.
- Downloaded RDP file contained internal FQDN.
- Remote Desktop Connection Broker generated Event ID 2056.
- Connection Broker database login failure prevented RemoteApp launch.

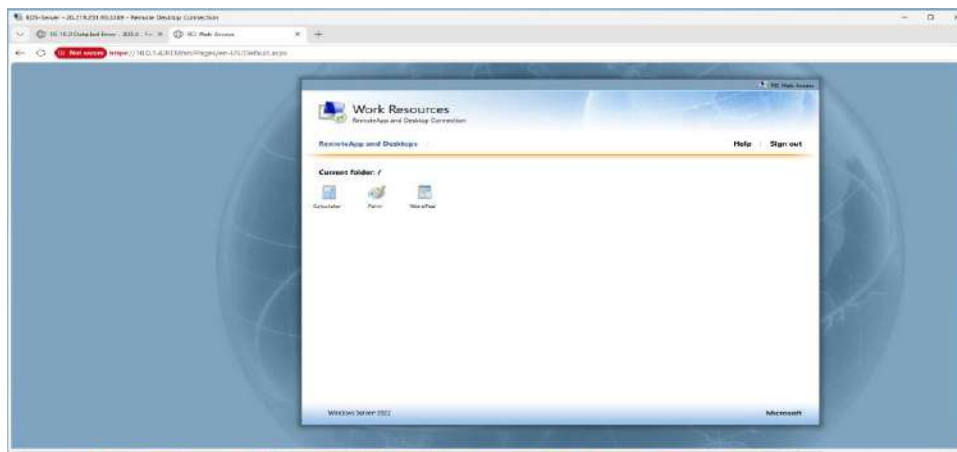
Root Cause:

The Remote Desktop Connection Broker database became inconsistent, causing RemoteApp files to reference the internal Active Directory name instead of the externally accessible Azure DNS name.

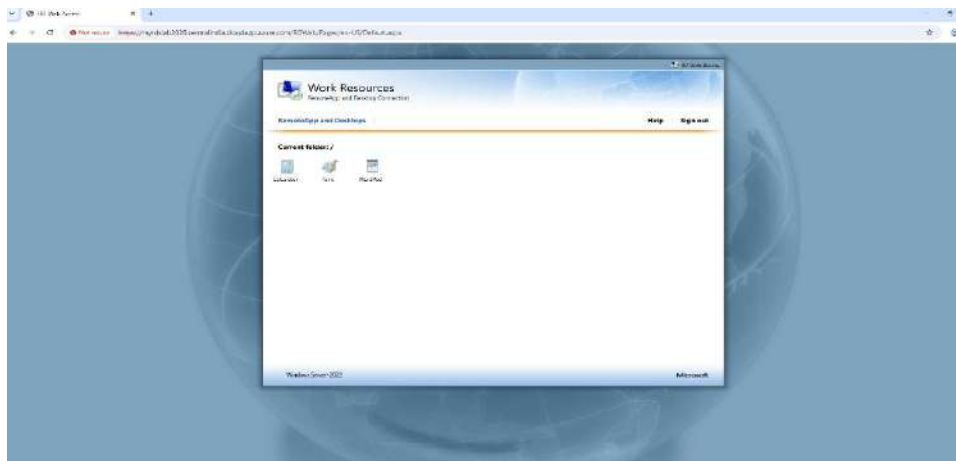
SS11. Proof of Delivery (Screenshots)



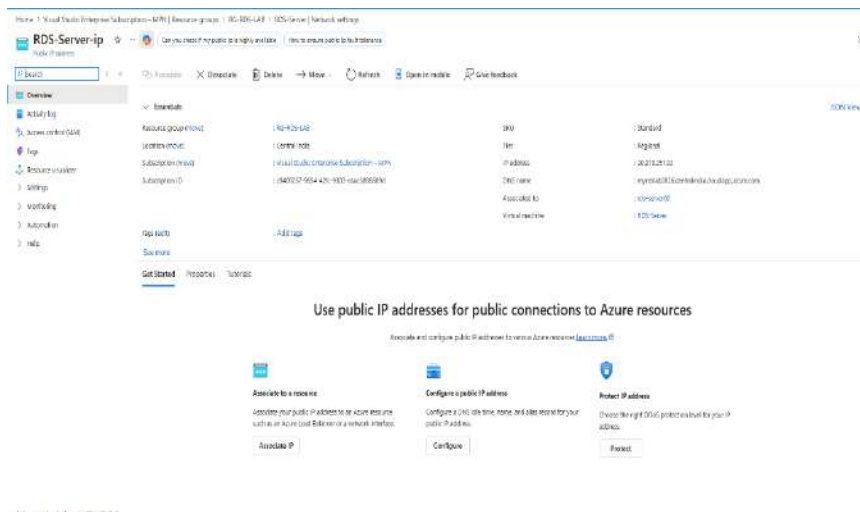
RD Web Access Portal



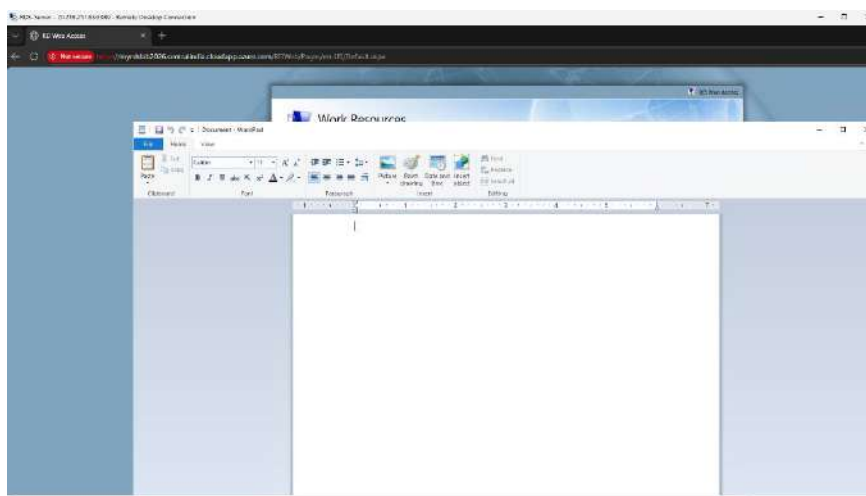
RD Web Access – Published RemoteApps



Azure Public IP Address Overview



Published RemoteApp (WordPad)



12. Conclusion

A complete Microsoft Azure Remote Desktop Services infrastructure was successfully deployed using a cost-optimized single-server architecture.

The environment included:

- RD Session Host
- RD Web Access
- RD Connection Broker
- Published RemoteApp

The RD Web portal functioned correctly and applications were published successfully.

During RemoteApp testing, a Connection Broker database inconsistency prevented external clients from resolving the published application endpoint. This issue was isolated to the RDS deployment configuration rather than Azure networking or RD Web Access.

The recommended corrective action is to recreate the RDS deployment or repair the Connection Broker database to ensure RemoteApp files reference the public DNS endpoint.